

29 January 2025

Polypropylene covalent adaptable networks with full cross-link density recovery after reprocessing: Development by free-radical reactive processing with resonance-stabilized, aromatic disulfide cross-linkers

Yen-Wen Huang², Mathew Suazo¹, Stephanie Barbon³, Hayley Brown³, Evelyn Auyeung³, Colin Li Pi Shan³, John Torkelson¹

1. Northwestern University

2. Northwestern University

3. The Dow Chemical Company

Abstract

A single-step method that produces percolated, dynamic covalent cross-links integrated into PP homopolymer has not been previously demonstrated. Here, we synthesized covalent adaptable networks (CANs) from polypropylene (PP) homopolymers using 180 °C, radical-based, reactive processing with a free-radical initiator, dicumyl peroxide (DCP), and resonance-stabilized, aromatic disulfide cross-linkers, one methacrylate-based and another phenylacrylate-based. Both cross-linkers yielded networks when reactively processed at 4 wt% with relatively high molecular weight (MW) PP (melt flow index (MFI) = 12) and 4 wt% DCP. The phenylacrylate-based cross-linker also yielded PP networks at other studied DCP/cross-linker concentrations and with relatively low MW PP (MFI = 35). Notably, our highest cross-link density PP CAN exhibited full recovery of cross-link density and tensile properties after three reprocessing steps by compression molding; that PP CAN also exhibited full cross-link density recovery within experimental uncertainty after reprocessing by melt extrusion.

Keywords

polypropylene, polyolefin, covalent adaptable network, vitrimer, free-radical, resonance stabilization, thermoset, disulfide, compression molding, melt extrusion, cross-link density, reprocessable, dynamic chemistry, peroxide, rubbery plateau modulus

**Polypropylene covalent adaptable networks with
full cross-link density recovery after reprocessing:
Development by free-radical reactive processing with
resonance-stabilized, aromatic disulfide cross-linkers**

Yen-Wen Huang,^a Mathew J. Suazo,^a Stephanie M. Barbon,^b Hayley A. Brown,^c

Evelyn Auyeung,^c Colin Li Pi Shan,^c and John M. Torkelson^{a,d,*}

^aDept. of Materials Science and Engineering, Northwestern University, Evanston, IL 60208, United States

^bThe Dow Chemical Company, Midland, MI 48764, United States

^cThe Dow Chemical Company, Lake Jackson, TX 77566, United States

^dDept. of Chemical and Biological Engineering, Northwestern University, Evanston, IL 60208, United States

*Corresponding author: j-torkelson@northwestern.edu

KEYWORDS: polypropylene; covalent adaptable network; dynamic chemistry; reactive processing; resonance stabilization

ABSTRACT: A single-step method that produces percolated, dynamic covalent cross-links integrated into PP homopolymer has not been previously demonstrated. Here, we synthesized covalent adaptable networks (CANs) from polypropylene (PP) homopolymers using 180 °C, radical-based, reactive processing with a free-radical initiator, dicumyl peroxide (DCP), and resonance-stabilized, aromatic disulfide cross-linkers, one methacrylate-based and another phenylacrylate-based. Both cross-linkers yielded networks when reactively processed at 4 wt% with relatively high molecular weight (MW) PP (melt flow index (MFI) = 12) and 4 wt% DCP. The phenylacrylate-based cross-linker also yielded PP networks at other studied DCP/cross-linker concentrations and with relatively low MW PP (MFI = 35). Notably, our highest cross-link density PP CAN exhibited full recovery of cross-link density and tensile properties after three reprocessing steps by compression molding; that PP CAN also exhibited full cross-link density recovery within experimental uncertainty after reprocessing by melt extrusion.

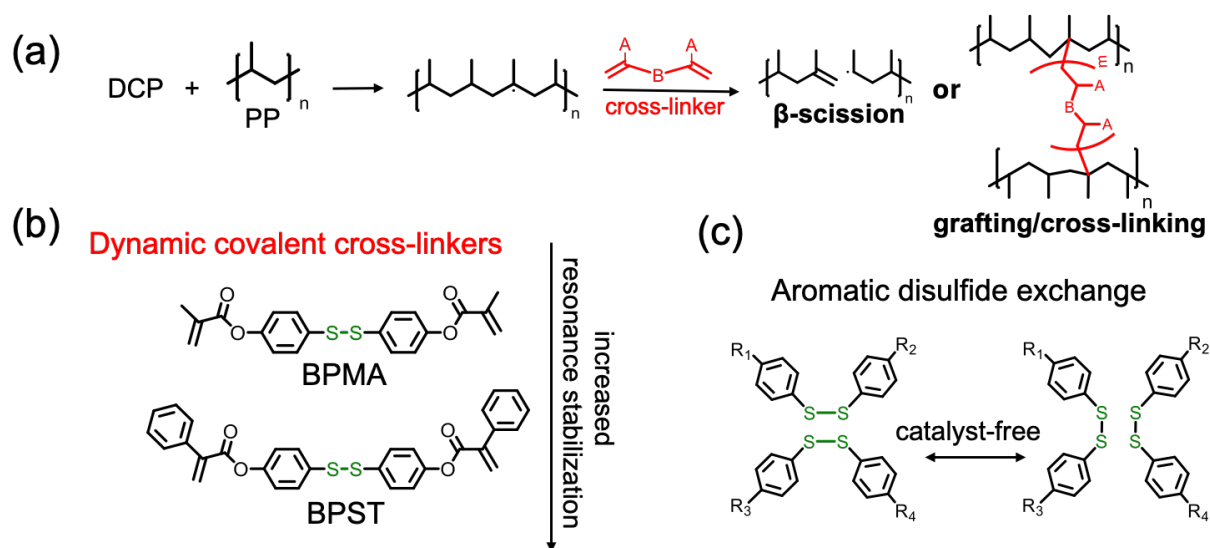
Polyethylene (PE) and polypropylene (PP) are prominent examples of polyolefins (POs),¹ ranking first (at 36 wt%) and second (at 21 wt%), respectively, in annual global production among all polymers.^{2,3} They possess favorable mechanical and insulation properties, excellent processability, and relatively low costs, making them useful in many applications, including films, containers, and packaging.⁴ Unfortunately, less than 10% of spent POs are recycled; most are landfilled or incinerated for energy, leading to significant environmental concerns.⁵ Common recycling methods for thermoplastic POs are melt extrusion and injection molding, which can reduce recycled product quality due to property degradation from repeated high-temperature processing.⁶ Such forms of thermoplastic recycling may result in downcycling, with the recycle used in lower-value applications.^{7,8}

Higher-value uses for PO thermoplastic waste have been investigated to create economic incentives for recycling, including converting thermoplastics into covalent adaptable networks (CANs),⁹ sometimes called vitrimers.^{10,11} Cross-linked PE (XLPE), made by the recombination of PE macroradicals generated by radical-based reactive processing or irradiation, shows property enhancements relative to thermoplastic PE, including improved thermal stability, chemical resistance, and impact strength. Consequently, XLPE is used in higher-value applications like foam, pipework, and cable insulation.¹² However, upcycling PE into XLPE has an environmental drawback: XLPE cannot be mechanically recycled, because the permanent covalent cross-links impede melt flow.¹³ To tackle the recyclability challenge with XLPE, advancements have been made in upcycling both virgin and post-consumer PE-based thermoplastics into reprocessable CANs.¹⁴⁻³² CANs feature dynamic covalent bonds that enable network reconfiguration under certain conditions, providing recyclability to cross-linked networks.^{9,11,33,34} Dynamic chemistries, such as transesterification,^{35,36} the Diels-Alder reaction,^{37,38} alkoxyamine chemistry,^{39,40} hindered urea exchange,^{41,42} and dialkylamino disulfide chemistry,^{43,44} along with those associated with urethanes,^{45,46} hydroxyurethanes,^{47,48} thiourethanes,^{49,50} and non-isocyanate thiourethanes,⁵¹ facilitate such reconfiguration at elevated temperatures. A significant breakthrough in the field of CANs is the development of catalyst-free PE CANs through a radical-based reactive process that

offers a simple, drop-in modification of a commercial method for producing XLPE.^{16,31,32} This drop-in approach involves adding peroxide and, as dynamic covalent cross-linkers, dimethacrylate-based dialkylamino or aromatic disulfides during radical-based reactive processing of ethylene-based thermoplastics.^{16-18,31,32,52} After the peroxide decomposes into primary free radicals (RO*) that abstract hydrogen atoms from PE, the resulting PE macroradicals can graft to the dynamic covalent cross-linkers or recombine to generate permanent cross-links.⁵³ By maintaining the level of permanent cross-links below the percolation limit while introducing percolated dynamic covalent cross-links,⁵⁴ the resulting CANs remain reprocessable.^{9,16,17}

Cross-linking of thermoplastic PP enhances specific properties: low levels of cross-links increase the PP melt strength and impact resistance at temperatures below 0 °C,⁵⁵ vital attributes for cross-linked PP foams.⁵⁶⁻⁵⁸ However, as shown in **Scheme 1a**, free radicals in PP prefer to react with the hydrogen atoms on tertiary carbons rather than those on primary and secondary carbons; the resulting tertiary macroradicals are prone to chain scission, rendering radical-based grafting and cross-linking ineffective.⁵⁹ Consequently, several studies have focused on generating cross-links through epoxy or thiol chemistry on pre-functionalized PP-g-MA (commercially produced by radical-based reactive processing with MA that causes major chain scission of the PP homopolymer⁶⁰), resulting in ester^{61,62} or thioester^{63,64} units that are dynamic with appropriate catalysts. One study⁶¹ reported the absence of a rubbery plateau modulus, the value of which is proportional to the network percolated cross-link density according to Flory's ideal rubber elasticity theory,⁶⁵ and average stress relaxation times below 0.5 s above the melt temperature; these outcomes are characteristic of flowable thermoplastics, not networks. A second study⁶⁴ reported that when extremely high MA levels (20-40 wt%) were used to produce the PP-g-MA starting material, reasonable rubbery plateau tensile moduli ($\cong 10^{-1}$ MPa) and gel contents (44-55%) for PP-g-MA network formation were achieved; the authors also noted that they “verified that several molding cycles did not significantly affect the material properties,”⁶⁴ but they provided no property characterization of (re)molded samples. A third study⁶² reported that PP-g-MA with standard (6 wt%) grafting levels could be processed into CANs with reasonable rubbery tensile

storage moduli and 42-58% gel fractions; reprocessability was demonstrated but with a loss in cross-link density and variations in apparent melting temperature with increasing (re)processing. Another study⁶³ converted brittle, low molecular weight (MW) PP-g-MA into a CAN and blended it at 15 wt% with 85 wt% high MW thermoplastic PP (PP_h). The blend exhibited properties comparable to those of neat PP_h and superior to those of a blend with 15 wt% thermoplastic, low MW PP-g-MA.



Scheme 1. (a) Free-radical-based approach for cross-linking conventional thermoplastic PP into PP networks, showing that β -scission is a common outcome that may prevent network formation; (b) aromatic disulfide dynamic covalent cross-linkers, with increasing resonance stabilization going from the methacrylate-based BPMA to the phenylacrylate-based BPST; (c) dynamic covalent character of aromatic disulfide cross-links at elevated temperature.

Notably, a 2024 study investigated single-step, peroxide-initiated vulcanization to establish sulfide-based dynamic linkages in a PP impact copolymer, which contains a low level of ethylene repeat units to enhance toughness.⁶⁶ Unfortunately, while ethylene units in the PP copolymer are expected to improve the effectiveness of radical-based processing compared to PP homopolymer, conventional characterizations of the rubbery plateau modulus and gel content showed no evidence

of cross-linking.⁶⁶ Furthermore, although the products of radical-based reactive processing exhibited tensile properties that differed from those of the original PP copolymer, reprocessing failed to yield recovery of properties within experimental uncertainty; such recovery is a desired trait in well-designed CANs. Importantly, a direct, single-step method that produces percolated, dynamic covalent cross-links integrated into PP homopolymer, which should be more challenging to achieve than with PP copolymer or PP-g-MA, has yet to be investigated.

Studies on grafting monomers like MA and glycidyl methacrylate onto PP indicate that adding styrene as a co-agent can suppress chain scission and enhance grafting yield.⁶⁷⁻⁷¹ As one of several potential factors contributing to this improved outcome, the resonance stabilization provided by the styrene phenyl group can delocalize the radical electron density, making PP less susceptible to chain scission.⁷¹ Divinylbenzene ($C_6H_4(CH=CH_2)_2$), which is related to styrene ($C_6H_5-CH=CH_2$) through the substitution of an aromatic hydrogen with a second vinyl group, has also been studied as a route to introduce permanent cross-links into PP.^{67,69,70,72} Here, inspired by the knowledge that a styrene co-agent can be beneficial in enhancing radical-based reactive grafting of PP, we compare the reactive radical-based processing of PP homopolymer into CANs using two aromatic disulfide-based dynamic cross-linkers (**Scheme 1b**): bis(4-methacryloyloxyphenyl) disulfide (BiPheS methacrylate, BPMA), previously used in making PE-based and other CANs,^{32,73} and bis(4-phenacryloyloxyphenyl) disulfide (BiPheS styrene, BPST), designed for this study. The resonance structures associated with BPMA are similarly associated with BPST. However, BPST has additional resonance structures from the phenyl ring attached to the carbon-carbon double bond, providing BPST with greater resonance stabilization than BPMA. The aromatic disulfide bonds in such cross-linkers have effective dynamic character at elevated temperatures;⁷⁴⁻⁷⁷ this may allow for reprocessability with cross-link density recovery of PP networks made successfully by this method.

We previously reported the BPMA synthesis procedure.⁶¹ The synthesis method for BPST is illustrated in **Scheme S1** and given in Supporting Information. The BPST synthesis was confirmed using 1H -NMR spectroscopy, ^{13}C -NMR spectroscopy, and electrospray ionization mass

spectroscopy (ESI-MS). (**Figures S1-S2**) Thermogravimetric analysis (TGA) (**Figure S3**) demonstrated high thermal stability for BPST ($T_{d,5\%} = 277\text{ }^{\circ}\text{C}$, similar to $T_{d,5\%} = 270\text{ }^{\circ}\text{C}$ for BPMA⁷³), more than sufficient thermostability for PP CAN melt-state (re)processing. **Figure S4** compares the Fourier-transform infrared (FT-IR) spectra of the two neat PPs used here with a commercially available PP homopolymer. The ratio of absorbance intensities associated with CH₂ (A_{1450}) and CH₃ (A_{1450}) units is the same within experimental uncertainty (**Table S1**), supporting the homopolymer nature of the PP samples.

The 1st mold PP CANs were prepared by reactively processing PP with dicumyl peroxide (DCP) and each dynamic cross-linker, following the procedures in Supporting Information and **Scheme S2**. Samples are named using the convention **PP type/cross-linker type wt %-DCP wt %**. For example, PP12/BPST8-4 denotes a sample created by reactively processing 8.0 wt% BPST and 4.0 wt% DCP with PP that has a melt flow index (MFI) of 12 (g/10 min, 230 °C/2.16kg; abbreviated as PP12). Samples were reactively processed at 180 °C for 15 min.

For the formulations listed in **Table 1**, the 1st mold samples of PP12/BPMA4-4 and PP12/BPST4-4 were prepared by reactively processing PP12 with 4.0 wt% DCP and 4.0 wt% BPMA or BPST. The network responses were examined by dynamic mechanical analysis (DMA) (**Figure 1a**). According to Flory's ideal rubber elasticity theory, the network rubbery plateau modulus is directly proportional to the effective cross-link density.⁶⁵ Additionally, a network with cross-links above the gel point should show a loss factor ($\tan \delta$) less than 1.⁷⁸ **Table 1** summarizes the tensile storage modulus, E' , and $\tan \delta$ values at 180 °C. PP12/BPMA4-4 and PP12/BPST4-4 exhibited significant rubbery plateaus, with E' values of 0.10 ± 0.02 and 0.12 ± 0.02 MPa, respectively, at 180 °C, and $\tan \delta < 0.4$. Network structures were also confirmed by gel contents (**Table 1**); PP12/BPMA4-4 and PP12/BPST4-4 have majority gel content (> 50%) whereas neat PP12 has 0% gel content. **Figure S5** shows that the gel content of PP12/BPST4-4 equals or exceeds 50% over a 170–190°C temperature range, indicating that the aromatic disulfide dynamic cross-linker design leads to networks with excellent integrity at elevated temperatures.

Table 1. Sample formulation, gel content, and DSC and DMA characterizations

Materials	[Cross-linker]-[DCP] (wt%) ^a	Gel content (%) ^b	$T_{m,peak}$ (°C) ^c	χ_c (%) ^d	E' @ 180 °C (MPa) ^e	$\tan \delta$ @ 180 °C ^e
PP35 ^f	0-0	0	163±1	32±2	<0.01	>1
PP35/BPMA4-4 ^f	4.0-4.0	30±4	155±1	23±2	<0.01	>1
PP35/BPST4-4 ^f	4.0-4.0	45±9	156±1	25±2	0.06±0.01	0.55±0.02
PP12 ^g	0-0	0	168±1	45±1	<0.01	>1
PP12/BPMA4-4 ^g	4.0-4.0	51±9	157±1	34±2	0.10±0.02	0.35±0.04
PP12/BPST4-4 ^g	4.0-4.0	53±9	160±1	34±2	0.12±0.02	0.30±0.03
PP12/BPMA4-1 ^g	4.0-1.0	13±2	161±1	33±2	<0.01	>1
PP12/BPST4-1 ^g	4.0-1.0	50±5	160±1	33±2	0.06±0.01	0.50±0.03
PP12/BPMA8-4 ^g	8.0-4.0	22±4	153±1	30±1	<0.01	>1
PP12/BPST8-4 ^g	8.0-4.0	67±15	153±1	28±1	0.27±0.04	0.20±0.04

^aWt% of cross-linkers and DCP used for reactive processing; the percentages of cross-linker and DCP were calculated based on the PP mass in the mixer.

^bObtained from samples swollen with *o*-xylene at 180°C.

^cDetermined from melting signal peaks obtained by DSC.

^dCrystallinity (χ_c) is calculated as the ratio of the melting heat of the PP samples to the enthalpy of melting for 100% crystalline PP (207.1 J/g).⁷⁹

^eDetermined by DMA.

^fPP precursor with MFI of 35 g/10 min at 230 °C under 2.16 kg load used.

^gPP precursor with MFI of 12 g/10 min at 230 °C under 2.16 kg load used.

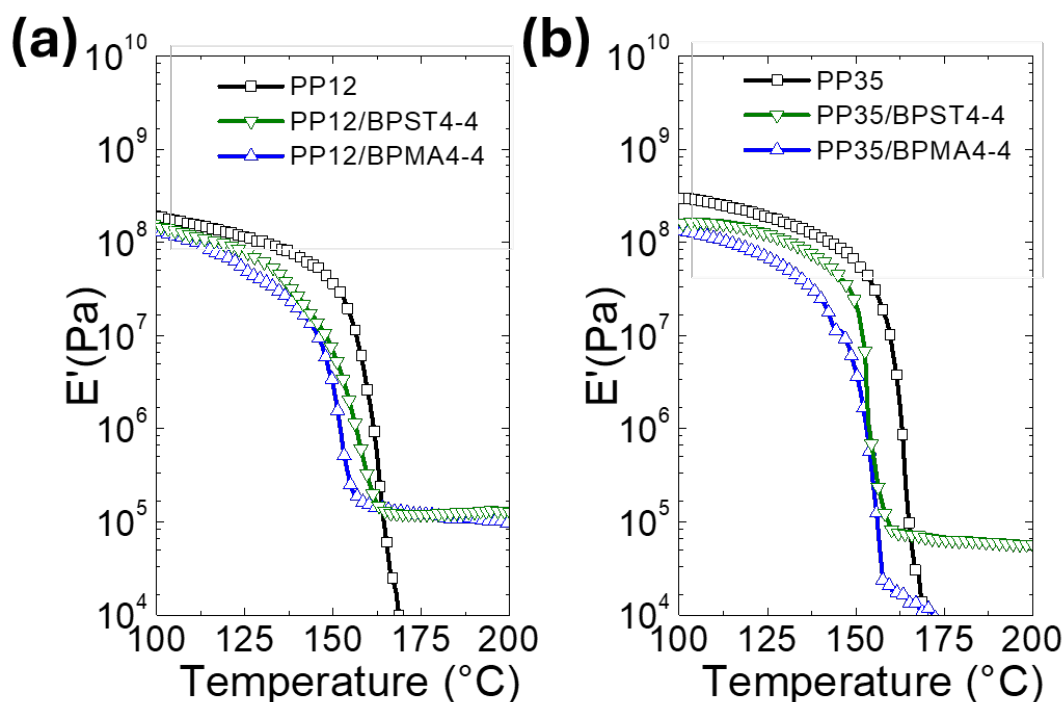


Figure 1. Tensile storage moduli (E') of neat PP and 1st mold of PP reactively processed at 180 °C for 15 min with 4.0 wt% DCP and 4.0 wt% BPMA or BPST. Results are presented for PP precursors with melt flow indexes (MFI) of (a) 12 (g/10 min, 230 °C/2.16 kg), and (b) 35 (g/10 min, 230 °C/2.16 kg). Legends read as “PP type”/“cross-linker type and wt %”-“DCP wt %”.

Differential scanning calorimetry (DSC) curves of the samples are presented in **Figure S6**. Given the semicrystalline nature of PP, after radical-based reactive processing, both chain scission and cross-linking hinder crystal formation during cooling, resulting in lamellae that are less thick than their neat precursors.¹⁷ This leads to reductions in crystallinity (χ_c) and melt transition peak temperature ($T_{m,peak}$) relative to neat PP. (See **Table 1** and **Figure S6**.) Given the low gel contents of PP12/BPMA4-1 and PP12/BPMA8-4, their crystallinity reductions can mainly be attributed to chain scission. For other samples with substantial gel contents, the decrease in crystallinity can be mainly attributed to hindered crystallization caused by cross-links.¹⁷

Lower MW, higher MFI PP requires a higher degree of grafting to cross-link successfully.

High MFI PP (35 g/10 min, 230 °C /2.16kg, abbreviated as PP35) was processed reactively with 4.0 wt% DCP and 4.0 wt% BPMA or BPST, resulting in 1st mold samples denoted as PP35/BPMA4-4 and PP35/BPST4-4, respectively. **Figure 1b** presents their E' curves, and **Table 1** summarizes their gel contents along with E' and $\tan \delta$ values at 180 °C. PP35/BPST4-4 showed a more significant rubbery plateau response ($E' = 0.06 \pm 0.01$ MPa at 180 °C) and gel content (45 ± 9 %) compared to PP35/BPMA4-4 ($E' < 0.01$ MPa at 180 °C; gel content of 30 ± 4 %), indicating that BPST is well suited to introduce dynamic covalent cross-links into PP35.

We investigated the effect of DCP loading (1.0 vs 4.0 wt%) with the PP12 samples by comparing 1st mold samples of PP12/BPMA4-4 and PP12/BPST4-4 with 1st mold samples of PP12/BPMA4-1 and PP12/BPST4-1. Due to fewer radicals being generated, samples made with 1.0 wt% DCP were less effectively cross-linked than those made with 4.0 wt% DCP.⁸⁰ **Figure 2a** and **Table 1** reveal the presence of a rubbery plateau response ($E' = 0.06 \pm 0.01$ MPa at 180 °C) and substantial 50 ± 5 % gel content in PP12/BPST4-1, whereas PP12/BPMA4-1 lacked a rubbery plateau and exhibited only 13 ± 2 % gel content. These results highlight the superiority of BPST over BPMA in enabling effective cross-linking of PP12 and PP35, even with 1.0 wt% DCP.

Varying the cross-linker loading is another strategy for controlling cross-link density. We compared PP12/BPMA4-4 and PP12/BPST4-4 with their counterparts containing a higher, 8.0 wt% cross-linker loading, namely PP12/BPMA8-4 and PP12/BPST8-4. **Figure 2b** and **Table 1** indicate that PP12/BPST8-4 exhibits higher rubbery plateau modulus (0.27 ± 0.04 MPa) and gel content (67 ± 15 %) than PP12/BPST4-4. However, unlike the cross-linked nature of PP12/BPMA4-4, flow was observed for PP12/BPMA8-4 at elevated temperatures, consistent with its relatively low gel content (22 ± 4 %). Similar behaviors have been noted in studies on grafting dynamic cross-linkers to PE and ethylene-based copolymers, attributed to the homopolymerization of dynamic cross-linkers at high initial concentrations, leading to inefficient grafting.¹⁶⁻¹⁸ We assessed the homopolymerization capability by polymerizing each cross-linker in a solvent with DCP as initiator. **Figure S7** reveals that BPMA undergoes homopolymerization to form

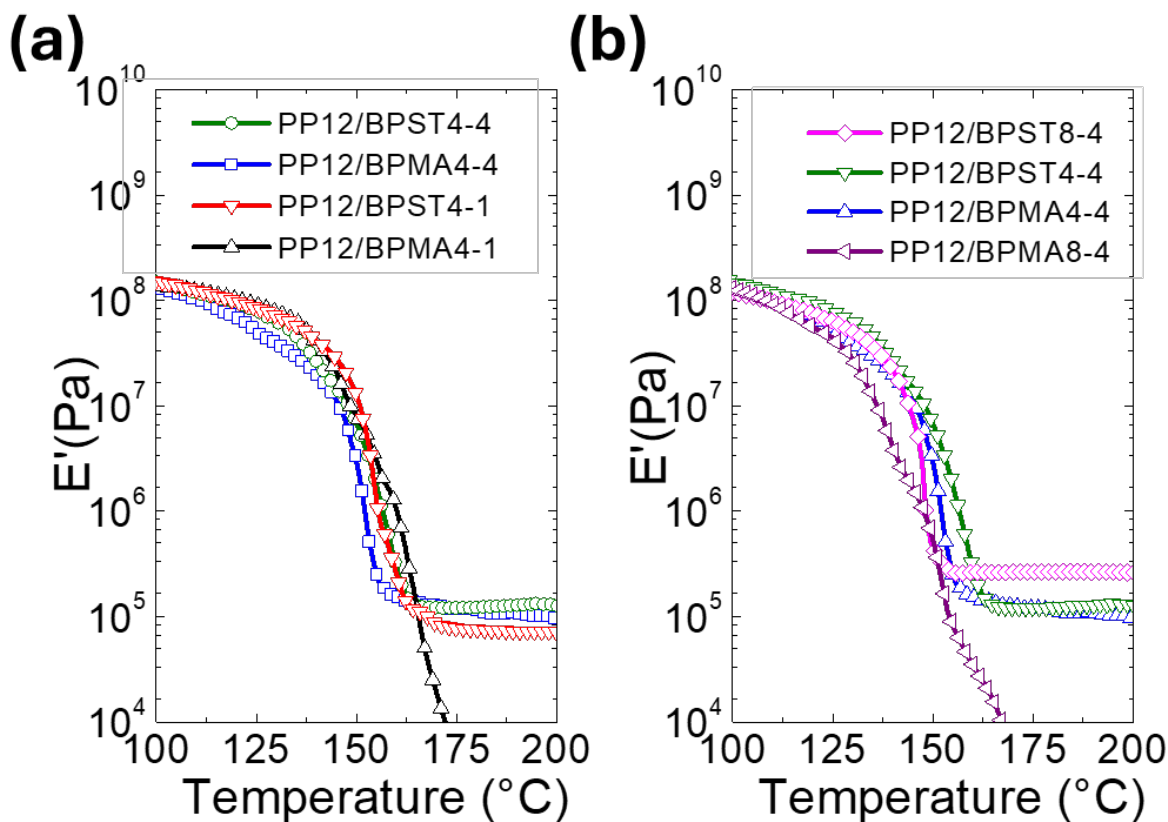


Figure 2. Tensile storage modulus (E') of neat PP (MFI = 12) and 1st mold of PP (MFI = 12) reactively processed with (a) 4.0 wt% BPMA or BPST with 1.0 or 4.0 wt% DCP, or (b) 4.0 wt% or 8.0 wt% of BPMA or BPST with 4.0 wt% DCP.

cross-linked gels. However, BPST remains uncross-linked after the reaction; the BPST phenyl substituent impedes homopolymerization, possibly due to greater resonance stabilization and/or steric hindrance.⁸¹⁻⁸⁵ This provides an additional benefit of BPST, allowing for facile tunability of the degree of cross-linking.

Figure S8 and **Table 2** compare the room-temperature tensile curves and tensile properties of PP12, PP12/BPST4-4, and PP12/BPST8-4. Notably, PP12/BPST8-4 exhibits Young's modulus, elongation-at-break, and toughness values that are the same within experimental uncertainty as those of PP12 and a value of elongation-at-yield that is double that of PP12. Only the tensile strength of PP12/BPST8-4 is reduced relative to PP12 (by ~25%). With a lower cross-link density,

Table 2. Room-temperature tensile test data for neat PP (MFI = 12 g/10 min, 230 °C/2.16 kg) and 1st mold samples of PP CANs made with 4.0 wt% or 8.0 wt% BPST and 4.0 wt% DCP.

Materials	Young's modulus ^a (MPa)	Tensile strength ^a (MPa)	Elongation- at-yield ^a (%)	Elongation- at-break ^a (%)	Toughness (MJ/m ³) ^a
PP12	800±70	27±3	6±1	250±40	60±7
PP12/BPST4-4	560±60	16±3	13±2	340±60	54±4
PP12/BPST8-4	720±50	20±2	12±2	300±40	60±5

^aError bars represent ± one standard deviation of three measurements. Tensile tests were done at a strain rate of 0.0038 s⁻¹.

PP12/BPST4-4 exhibits values of elongation-at-break and toughness similar to those of PP12 but substantially reduced values of Young's modulus (by ~30%) and tensile strength (by ~40%). Like PP12/BPST8-4, PP12/BPST4-4 exhibits an elongation-at-yield about twice that of PP12.

The PP12/BPST8-4 CAN was chosen for reprocessing because it exhibited the highest rubbery plateau E' and gel content of all CAN samples (**Table 1**). We assessed the reprocessability of PP12/BPST8-4 by remolding 1st mold samples (see Supporting Information for procedures) into 2nd mold and 3rd mold samples. **Figure 3a** shows that a consolidated PP12/BPST8-4 film can be recovered from cut pieces by compression molding for 10 min at 180 °C, which is lower than typical PP melt-processing temperatures (often ≥ 200 °C).⁸⁶ **Figure 3b** presents the temperature-dependent E' responses of the 1st, 2nd, and 3rd mold samples; **Table S2** summarizes the E' values at 180 °C. The 2nd and 3rd mold samples show rubbery plateau E' responses that are identical within experimental uncertainty to that of the 1st mold sample, thus confirming full cross-link density recovery. DSC and FT-IR spectroscopy characterizations of 1st, 2nd, and 3rd mold samples are displayed in **Figure S9** and summarized in **Table S2**. These findings further confirm the excellent

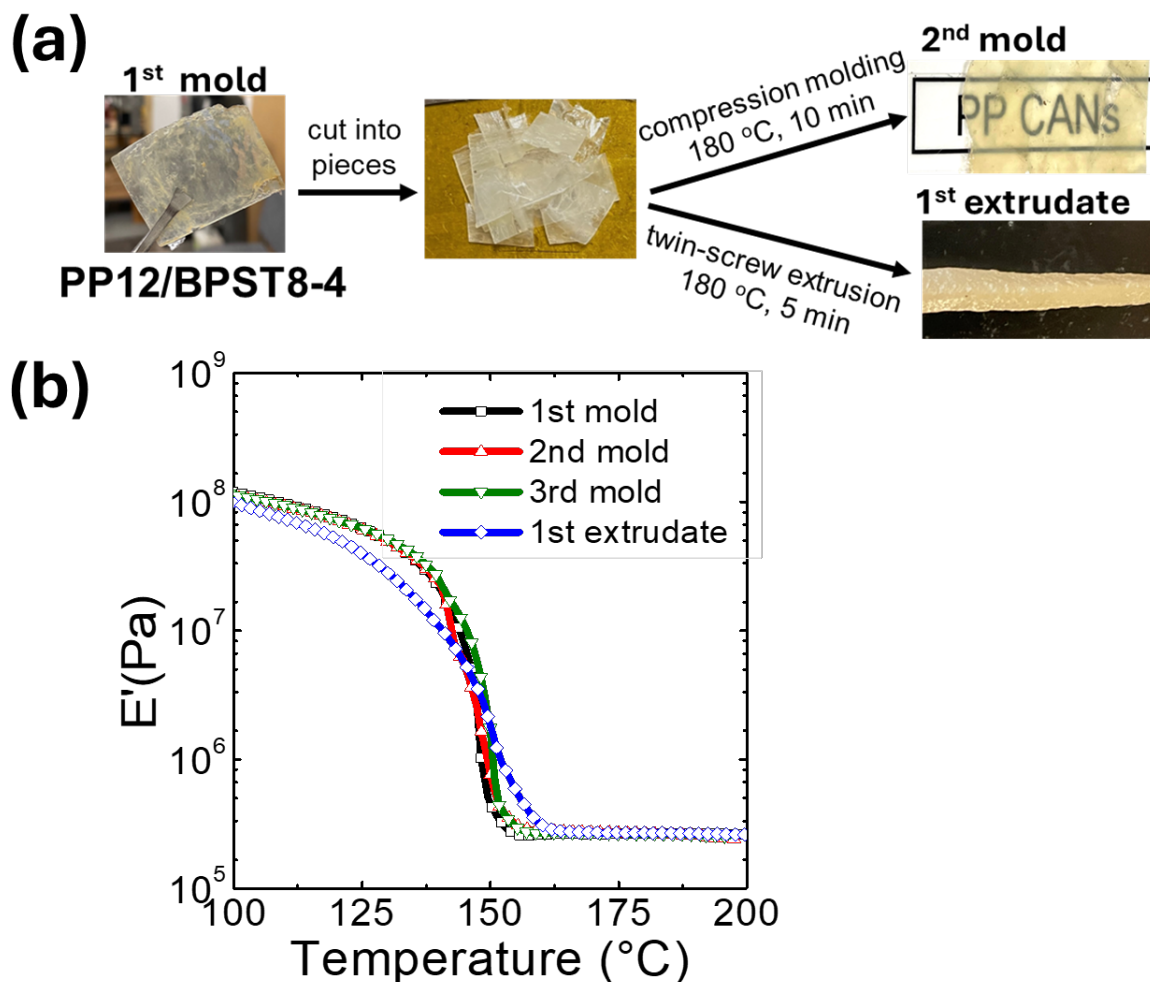


Figure 3. (a) Images of 2nd mold and 1st extrudate reprocessed from cut pieces of 1st mold samples of PP12/BPST8-4; (b) tensile storage modulus (E') of 1st, 2nd, and 3rd mold samples as well as 1st extrudate of PP12/BPST8-4.

property recovery of the PP12/BPST8-4 CAN after reprocessing.

We also demonstrate the melt extrudability of our PP CANs. **Figures 3a** and **S10** show that 1st mold samples of PP12/BPST8-4 can be extruded using a HAAKE Minilab 3 extruder to produce a sample referred to as the 1st extrudate. The DMA profile for the 1st extrudate (**Figure 3b**) shows that the rubbery plateau E' values overlap well with those of the (re)molded samples. The E' value

at 180 °C of 0.22 ± 0.05 MPa for the extrudate is, within experimental uncertainty, identical to those of the (re)molded samples (**Table 1; Table S2**) and is consistent with recovery of cross-link density after reprocessing by extrusion.

In summary, we prepared the first CANs produced in one step from PP homopolymer by introducing dynamic covalent cross-links via radical-based reactive processing with the radical initiator DCP and the aromatic disulfide cross-linkers BPMA and BPST. Using relatively high MW PP (MFI of 12), we demonstrated that PP networks may be created by reactive processing at 180 °C with 4 wt% DCP and 4 wt % BPMA or BPST. However, only samples produced with BPST resulted in significant network formation when reactively processed with relatively low MW PP (MFI of 35) and 4 wt% each of DCP and BPMA or BPST or when reactively processed with relatively high MW PP and 1 wt% DCP and 4 wt% cross-linker loadings or 4 wt% DCP and 8 wt% cross-linker loadings. All our attempts at developing PP networks using BPST, which is designed with stronger resonance stabilization than BPMA and may afford better solubility in nonpolar PP than BPMA, resulted in network formation. We also demonstrated full recovery of cross-link density and tensile properties within experimental uncertainty in a BPST-based PP CAN after reprocessing. These outcomes show the potential of this approach for upcycling thermoplastic PP, whether virgin or waste, into reprocessable PP CANs.

Supporting Information. Materials and experimental procedures; ^1H -NMR, ^{13}C -NMR, ESI-MS spectra and TGA of BPST; network preparation; FT-IR and DSC curves of samples; homopolymerization of dynamic covalent cross-linkers; tensile properties of samples; properties after CAN reprocessing; demonstration of twin-screw extrusion.

Conflicts of Interest. A provisional US patent application related to this work has been filed.

Acknowledgments. This research was supported by the University Partnership Initiative between Northwestern University and The Dow Chemical Company. We also gratefully acknowledge the

support from Northwestern University via discretionary funds associated with a Walter P. Murphy Professorship (J.M.T.). This work made use of the MatCI Facility at Northwestern University, which receives support from the MRSEC Program (NSFDMR-1720139) of the Materials Research Center at Northwestern University.

References

- (1) Galli, P.; Vecellio, G. Technology: driving force behind innovation and growth of polyolefins. *Prog. Polym. Sci.* **2001**, *26* (8), 1287-1336.
- (2) Geyer, R.; Jambeck, J. R.; Law, K. L. Production, use, and fate of all plastics ever made. *Sci. Adv.* **2017**, *3* (7), e1700782.
- (3) Heller, M. C.; Mazor, M. H.; Keoleian, G. A. Plastics in the US: toward a material flow characterization of production, markets and end of life. *Environ. Res. Lett.* **2020**, *15* (9), 094034.
- (4) Chum, P. S.; Swogger, K. W. Olefin polymer technologies—History and recent progress at The Dow Chemical Company. *Prog. Polym. Sci.* **2008**, *33* (8), 797-819.
- (5) Zhao, X.; Korey, M.; Li, K.; Copenhaver, K.; Tekinalp, H.; Celik, S.; Kalaitzidou, K.; Ruan, R.; Ragauskas, A. J.; Ozcan, S. Plastic waste upcycling toward a circular economy. *Chem. Eng. J.* **2022**, *428*, 131928.
- (6) Yin, S.; Tuladhar, R.; Shi, F.; Shanks, R. A.; Combe, M.; Collister, T. Mechanical reprocessing of polyolefin waste: A review. *Polym. Eng. Sci.* **2015**, *55* (12), 2899-2909.
- (7) Achilias, D.; Antonakou, E.; Roupakias, C.; Megalokonomos, P.; Lappas, A. Recycling techniques of polyolefins from plastic wastes. *Global NEST J.* **2008**, *10* (1), 114-122.
- (8) Jubinville, D.; Esmizadeh, E.; Saikrishnan, S.; Tzoganakis, C.; Mekonnen, T. A comprehensive review of global production and recycling methods of polyolefin (PO) based products and their post-recycling applications. *Sustain. Mater. Techno.* **2020**, *25*, e00188.
- (9) Kloxin, C. J.; Scott, T. F.; Adzima, B. J.; Bowman, C. N. Covalent adaptable networks (CANs): a unique paradigm in cross-linked polymers. *Macromolecules* **2010**, *43* (6), 2643-2653.
- (10) Capelot, M.; Unterlass, M. M.; Tournilhac, F.; Leibler, L. Catalytic control of the vitrimer

glass transition. *ACS Macro Lett.* **2012**, *1* (7), 789-792.

(11) Denissen, W.; Winne, J. M.; Du Prez, F. E. Vitrimers: permanent organic networks with glass-like fluidity. *Chem. Sci.* **2016**, *7* (1), 30-38.

(12) Ahmad, H.; Rodrigue, D. Crosslinked polyethylene: A review on the crosslinking techniques, manufacturing methods, applications, and recycling. *Polym. Eng. Sci.* **2022**, *62* (8), 2376-2401.

(13) Selvin, M.; Shah, S.; Maria, H. J.; Thomas, S.; Tuladhar, R.; Jacob, M. Review on Recycling of Cross-Linked Polyethylene. *Ind. Eng. Chem. Res.* **2024**, *63* (3), 1200-1214.

(14) Boen, N. K.; Hillmyer, M. A. Post-polymerization functionalization of polyolefins. *Chem. Soc. Rev.* **2005**, *34* (3), 267-275.

(15) Navarre, S.; Degueil, M.; Maillard, B. Chemical modification of molten polyethylene by thermolysis of peroxyketals. *Polymer* **2001**, *42* (10), 4509-4516.

(16) Fenimore, L. M.; Chen, B.; Torkelson, J. M. Simple upcycling of virgin and waste polyethylene into covalent adaptable networks: catalyst-free, radical-based reactive processing with dialkylamino disulfide bonds. *J. Mater. Chem. A* **2022**, *10* (46), 24726-24745.

(17) Fenimore, L. M.; Chen, B.; Chen, Y.; Barbon, S. M.; Brown, H. A.; Auyeung, E.; Shan, C. L. P.; Torkelson, J. M. Covalent adaptable networks and thermosets of multi-block ethylene/1-octene copolymers made by free-radical processing: Effects of melt flow index and crystallinity on thermomechanical properties and reprocessability. *Eur. Polym. J.* **2024**, *202*, 112661.

(18) Chen, B.; Fenimore, L. M.; Chen, Y.; Barbon, S. M.; Brown, H. A.; Auyeung, E.; Shan, C. L. P.; Torkelson, J. M. Novel covalent adaptable networks (CANs) of ethylene/1-octene copolymers (EOCs) made by free-radical processing: comparison of structure–property relationships of EOC CANs with EOC thermosets. *Polym. Chem.* **2023**, *14* (31), 3621-3637.

(19) Polgar, L.; Van Duin, M.; Broekhuis, A.; Picchioni, F. Use of Diels–Alder chemistry for thermoreversible cross-linking of rubbers: the next step toward recycling of rubber products? *Macromolecules* **2015**, *48* (19), 7096-7105.

(20) Röttger, M.; Domenech, T.; van Der Weegen, R.; Breuillac, A.; Nicolaÿ, R.; Leibler, L.

High-performance vitrimers from commodity thermoplastics through dioxaborolane metathesis. *Science* **2017**, *356* (6333), 62-65.

(21) Ji, F.; Liu, X.; Lin, C.; Zhou, Y.; Dong, L.; Xu, S.; Sheng, D.; Yang, Y. Reprocessable and recyclable crosslinked polyethylene with triple shape memory effect. *Macro. Mater. Eng.* **2019**, *304* (3), 1800528.

(22) Caffy, F.; Nicolaÿ, R. Transformation of polyethylene into a vitrimer by nitroxide radical coupling of a bis-dioxaborolane. *Polym. Chem.* **2019**, *10* (23), 3107-3115.

(23) Ricarte, R. G.; Tournilhac, F.; Leibler, L. Phase separation and self-assembly in vitrimers: Hierarchical morphology of molten and semicrystalline polyethylene/dioxaborolane maleimide systems. *Macromolecules* **2018**, *52* (2), 432-443.

(24) Tellers, J.; Pinalli, R.; Soliman, M.; Vachon, J.; Dalcanale, E. Reprocessable vinylogous urethane cross-linked polyethylene via reactive extrusion. *Polym. Chem.* **2019**, *10* (40), 5534-5542.

(25) Wang, Z.; Gu, Y.; Ma, M.; Liu, Y.; Chen, M. Strengthening polyethylene thermoplastics through a dynamic covalent networking additive based on alkylboron chemistry. *Macromolecules* **2021**, *54* (4), 1760-1766.

(26) Wang, S.; Ma, S.; Qiu, J.; Tian, A.; Li, Q.; Xu, X.; Wang, B.; Lu, N.; Liu, Y.; Zhu, J. Upcycling of post-consumer polyolefin plastics to covalent adaptable networks via in situ continuous extrusion cross-linking. *Green Chem.* **2021**, *23* (8), 2931-2937.

(27) Liu, Y.; Tang, Z.; Chen, J.; Xiong, J.; Wang, D.; Wang, S.; Wu, S.; Guo, B. Tuning the mechanical and dynamic properties of imine bond crosslinked elastomeric vitrimers by manipulating the crosslinking degree. *Polym. Chem.* **2020**, *11* (7), 1348-1355.

(28) Yang, F.; Pan, L.; Ma, Z.; Lou, Y.; Li, Y.; Li, Y. Highly elastic, strong, and reprocessable cross-linked polyolefin elastomers enabled by boronic ester bonds. *Polym. Chem.* **2020**, *11* (19), 3285-3295.

(29) Huang, Q.; Tang, Z.; Wang, D.; Wu, S.; Guo, B. Engineering segregated structures in a cross-linked elastomeric network enabled by dynamic cross-link reshuffling. *ACS Macro Lett.*

2021, *10* (2), 231-236.

(30) Ahmadi, M.; Hanifpour, A.; Ghiassinejad, S.; van Ruymbeke, E. Polyolefins vitrimers: design principles and applications. *Chem. Mater.* **2022**, *34* (23), 10249-10271.

(31) Chen, B.; Debsharma, T.; Fenimore, L. M.; Wang, T.; Chen, Y.; Purwanto, N. S.; Torkelson, J. M. Rapidly Self-Healable and Melt-Extrudable Polyethylene Reprocessable Network Enabled with Dialkylamino Disulfide Dynamic Chemistry. *Macromol. Rapid Commun.* **2024**, *45* (22), 2400460.

(32) Suazo, M. J.; Fenimore, L. M.; Barbon, S. M.; Brown, H.; Auyeung, E.; Cespedes, G.; Li Pi Shan, C.; Torkelson, J. M. Extrudable and Highly Creep-Resistant Covalent Adaptable Networks Made from Polyethylene and Ethylene/1-Octene Copolymers by Reactive Processing with Aromatic Disulfide Cross-Links. *ACS Appl. Polym. Mater.* **2024**, *6* (23), 14772-14783.

(33) Zheng, N.; Xu, Y.; Zhao, Q.; Xie, T. Dynamic covalent polymer networks: a molecular platform for designing functions beyond chemical recycling and self-healing. *Chem. Rev.* **2021**, *121* (3), 1716-1745.

(34) Krishnakumar, B.; Sanka, R. P.; Binder, W. H.; Parthasarthy, V.; Rana, S.; Karak, N. Vitrimers: Associative dynamic covalent adaptive networks in thermoset polymers. *Chem. Eng. J.* **2020**, *385*, 123820.

(35) Schenk, V.; Labastie, K.; Destarac, M.; Olivier, P.; Guerre, M. Vitrimer composites: Current state and future challenges. *Mater. Adv.* **2022**, *3* (22), 8012-8029.

(36) Guerre, M.; Taplan, C.; Winne, J. M.; Du Prez, F. E. Vitrimers: directing chemical reactivity to control material properties. *Chem. Sci.* **2020**, *11* (19), 4855-4870.

(37) Ratwani, C. R.; Kamali, A. R.; Abdelkader, A. M. Self-healing by Diels-Alder cycloaddition in advanced functional polymers: A review. *Prog. Mater. Sci.* **2023**, *131*, 101001.

(38) Briou, B.; Améduri, B.; Boutevin, B. Trends in the Diels–Alder reaction in polymer chemistry. *Chem. Soc. Rev.* **2021**, *50* (19), 11055-11097.

(39) Li, L.; Chen, X.; Jin, K.; Bin Rusayyis, M.; Torkelson, J. M. Arresting elevated-temperature creep and achieving full cross-link density recovery in reprocessable polymer networks and

network composites via nitroxide-mediated dynamic chemistry. *Macromolecules* **2021**, *54* (3), 1452-1464.

(40) Jin, K.; Li, L.; Torkelson, J. M. Recyclable crosslinked polymer networks via one-step controlled radical polymerization. *Adv. Mater.* **2016**, *28* (31), 6746-6750.

(41) Zhang, Q.; Wang, S.; Rao, B.; Chen, X.; Ma, L.; Cui, C.; Zhong, Q.; Li, Z.; Cheng, Y.; Zhang, Y. Hindered urea bonds for dynamic polymers: An overview. *React. Funct. Polym.* **2021**, *159*, 104807.

(42) Ying, H.; Zhang, Y.; Cheng, J. Dynamic urea bond for the design of reversible and self-healing polymers. *Nat. Commun.* **2014**, *5* (1), 3218.

(43) Bin Rusayyis, M.; Torkelson, J. M. Reprocessable covalent adaptable networks with excellent elevated-temperature creep resistance: facilitation by dynamic, dissociative bis (hindered amino) disulfide bonds. *Polym. Chem.* **2021**, *12* (18), 2760-2771.

(44) Bin Rusayyis, M.; Torkelson, J. M. Recyclable polymethacrylate networks containing dynamic dialkylamino disulfide linkages and exhibiting full property recovery. *Macromolecules* **2020**, *53* (19), 8367-8373.

(45) Fortman, D. J.; Sheppard, D. T.; Dichtel, W. R. Reprocessing cross-linked polyurethanes by catalyzing carbamate exchange. *Macromolecules* **2019**, *52* (16), 6330-6335.

(46) Chen, X.; Hu, S.; Li, L.; Torkelson, J. M. Dynamic covalent polyurethane networks with excellent property and cross-link density recovery after recycling and potential for monomer recovery. *ACS Appl. Polym. Mater.* **2020**, *2* (5), 2093-2101.

(47) Fortman, D. J.; Brutman, J. P.; Cramer, C. J.; Hillmyer, M. A.; Dichtel, W. R. Mechanically activated, catalyst-free polyhydroxyurethane vitrimers. *J. Am. Chem. Soc.* **2015**, *137* (44), 14019-14022.

(48) Chen, X.; Li, L.; Jin, K.; Torkelson, J. M. Reprocessable polyhydroxyurethane networks exhibiting full property recovery and concurrent associative and dissociative dynamic chemistry via transcarbamoylation and reversible cyclic carbonate aminolysis. *Polym. Chem.* **2017**, *8* (41), 6349-6355.

- (49) Cui, C.; Chen, X.; Ma, L.; Zhong, Q.; Li, Z.; Mariappan, A.; Zhang, Q.; Cheng, Y.; He, G.; Chen, X. Polythiourethane covalent adaptable networks for strong and reworkable adhesives and fully recyclable carbon fiber-reinforced composites. *ACS Appl. Mater. Inter.* **2020**, *12* (42), 47975-47983.
- (50) Li, L.; Chen, X.; Torkelson, J. M. Reprocessable polymer networks via thiourethane dynamic chemistry: recovery of cross-link density after recycling and proof-of-principle solvolysis leading to monomer recovery. *Macromolecules* **2019**, *52* (21), 8207-8216.
- (51) Chen, Y.; Chen, B.; Torkelson, J. M. Biobased, reprocessable non-isocyanate polythiourethane networks with thionourethane and disulfide cross-links: Comparison with polyhydroxyurethane network analogues. *Macromolecules* **2023**, *56* (10), 3687-3702.
- (52) Fenimore, L. M.; Suazo, M. J.; Torkelson, J. M. Covalent Adaptable Networks Made by Reactive Processing of Highly Entangled Polymer: Synthesis-Structure-Thermomechanical Property-Reprocessing Relationship in Covalent Adaptable Networks. *Macromolecules* **2024**, *57* (6), 2756-2772.
- (53) Li, T.-T.; Cheng, S.-B.; Feng, L.-F.; Gu, X.-P.; Duan, J.-T.; Jiang, M.-Z.; Zhang, C.-L. A review on the free radical grafting of vinyl monomers onto polyethylene and polypropylene by reactive extrusion. *Chem. Eng. Sci.* **2023**, 118916.
- (54) Li, L.; Chen, X.; Jin, K.; Torkelson, J. M. Vitrimers designed both to strongly suppress creep and to recover original cross-link density after reprocessing: quantitative theory and experiments. *Macromolecules* **2018**, *51* (15), 5537-5546.
- (55) Rätzsch, M.; Arnold, M.; Borsig, E.; Bucka, H.; Reichelt, N. Radical reactions on polypropylene in the solid state. *Prog. Polym. Sci.* **2002**, *27* (7), 1195-1282.
- (56) Liu, F.; Shen, C.; You, F.; Zhao, W.; Deng, C.; Jiang, X. Enhancing the sound and thermal insulation properties of polypropylene foam by preparing high melt strength polypropylene. *Macromol. Rapid Commun.* **2023**, *44* (20), 2300344.
- (57) Yeh, S. K.; Tsai, Y. B.; Gebremedhin, K. F.; Chien, T. Y.; Chang, R. Y.; Tung, K. L. Preparation of polypropylene/high-melt-strength PP open-cell foam for oil absorption. *Polym.*

Eng. Sci. **2021**, *61* (4), 1139-1149.

(58) Mrlik, M.; Al Maadeed, M. A. A. Tailoring of the thermal, mechanical and dielectric properties of the polypropylene foams using gamma-irradiation. *Polym. Degrad. Stab.* **2016**, *133*, 234-242.

(59) Moad, G. The synthesis of polyolefin graft copolymers by reactive extrusion. *Prog. Polym. Sci.* **1999**, *24* (1), 81-142.

(60) Diop, M. F.; Torkelson, J. M. Maleic anhydride functionalization of polypropylene with suppressed molecular weight reduction via solid-state shear pulverization. *Polymer* **2013**, *54* (16), 4143-4154.

(61) Gao, Y.; Niu, H. Polypropylene-based transesterification covalent adaptable networks with internal catalysis. *Polym. Chem.* **2024**, *15* (9), 884-895.

(62) Kar, G. P.; Saed, M. O.; Terentjev, E. M. Scalable upcycling of thermoplastic polyolefins into vitrimers through transesterification. *J. Mater. Chem. A* **2020**, *8* (45), 24137-24147.

(63) Sadri, M.; Patil, S.; Perkins, J.; Gunter, Z.; Cheng, S.; Qiang, Z. Polymeric Dynamic Crosslinker for Upcycling of Fragile Low-Molecular-Weight Polypropylene. *ACS Appl. Polym. Mater.* **2023**, *5* (6), 4056-4068.

(64) Saed, M. O.; Lin, X.; Terentjev, E. M. Dynamic semicrystalline networks of polypropylene with thiol-anhydride exchangeable crosslinks. *ACS Appl. Mater. Inter.* **2021**, *13* (35), 42044-42051.

(65) Flory, P. J. *Principles of Polymer Chemistry*; Cornell University Press, 1953.

(66) Houghton-Flory, C.; Saed, M. O.; Terentjev, E. M. Vulcanization of polypropylene. *J. Polym. Sci.* **2024**, *62* (9), 1902-1914.

(67) Gu, J.; Xu, H.; Wu, C. The effect of benzoyl peroxide and divinyl benzene on the properties of cross-linked recycled polyolefin blends. *J. Macromol. Sci. B* **2014**, *53* (12), 1777-1785.

(68) Li, Y.; Xie, X.-M.; Guo, B.-H. Study on styrene-assisted melt free-radical grafting of maleic anhydride onto polypropylene. *Polymer* **2001**, *42* (8), 3419-3425.

(69) Hou, Y.-J.; Xie, B.-H.; Yang, W.; Yang, M.-B. Micro-structure and fracture behavior of high-

melt-strength PPs prepared by reactive extrusion. *J. Macromol. Sci. B* **2012**, *51* (1), 48-59.

(70) Zhao, J.; Chen, M.; Wang, X.; Zhao, X.; Wang, Z.; Dang, Z.-M.; Ma, L.; Hu, G.-H.; Chen, F. Triple shape memory effects of cross-linked polyethylene/polypropylene blends with cocontinuous architecture. *ACS Appl. Mater. Inter.* **2013**, *5* (12), 5550-5556.

(71) Hu, G. H.; Cartier, H. Styrene-assisted melt free radical grafting of glycidyl methacrylate onto an ethylene and propylene rubber. *J. Appl. Polym. Sci.* **1999**, *71* (1), 125-133.

(72) Yang, M.; Li, J.; Guo, S. A reactive extrusion process with the aid of ultrasound for preparing cross-linked polypropylene. *Polym. Eng. Sci.* **2017**, *57* (8), 821-829.

(73) Suazo, M. J.; Torkelson, J. M. Reprocessable covalent adaptable networks via free-radical polymerization with an aromatic disulfide cross-linker. *ACS Appl. Polym. Mater.* **2024**, *6* (15), 9209-9218.

(74) Azcune, I.; Odriozola, I. Aromatic disulfide crosslinks in polymer systems: Self-healing, reprocessability, recyclability and more. *Eur. Polym. J.* **2016**, *84*, 147-160.

(75) Xiang, S.; Zhou, L.; Chen, R.; Zhang, K.; Chen, M. Interlocked covalent adaptable networks and composites relying on parallel connection of aromatic disulfide and aromatic imine cross-links in epoxy. *Macromolecules* **2022**, *55* (23), 10276-10284.

(76) Shan, S.; Mai, D.; Lin, Y.; Zhang, A. Self-healing, reprocessable, and degradable bio-based epoxy elastomer bearing aromatic disulfide bonds and its application in strain sensors. *ACS Appl. Polym. Mater.* **2021**, *3* (10), 5115-5124.

(77) Li, T.; Xie, Z.; Xu, J.; Weng, Y.; Guo, B.-H. Design of a self-healing cross-linked polyurea with dynamic cross-links based on disulfide bonds and hydrogen bonding. *Eur. Polym. J.* **2018**, *107*, 249-257.

(78) Purwanto, N. S.; Chen, Y.; Wang, T.; Torkelson, J. M. Rapidly synthesized, self-blowing, non-isocyanate polyurethane network foams with reprocessing to bulk networks via hydroxyurethane dynamic chemistry. *Polymer* **2023**, *272*, 125858.

(79) Bu, H. S.; Cheng, S. Z.; Wunderlich, B. Addendum to the thermal properties of polypropylene. *Makromol. Chem., Rapid Commun.* **1988**, *9* (2), 75-77.

- (80) Wang, D.; Xie, X. M.; Jow, J.; Chen, H. Y.; Lai, S. Y. Styrene-assisted melt free-radical grafting of pentaerythritol triacrylate onto polypropylene and its crystallization behavior. *J. Appl. Polym. Sci.* **2008**, *108* (3), 1737-1743.
- (81) Braun, D.; Hu, F. Polymers from non-homopolymerizable monomers by free radical processes. *Prog. Polym. Sci.* **2006**, *31* (3), 239-276.
- (82) Kohsaka, Y.; Akae, Y.; Kawatani, R.; Kazama, A. Polymer chemistry of α -substituted acrylates designed for functional-group synergy. *J. Macromol. Sci. Chem., A* **2022**, *59* (2), 83-97.
- (83) Wan, D.; Shi, L.; Huang, J. Spontaneous alternating copolymerization of N-phenylmaleimide with ethyl α -phenylacrylate. *J. Polym. Sci., Part A: Polym. Chem.* **1998**, *36* (16), 2927-2931.
- (84) Yuki, H.; Ohta, K.; Hatada, K.; Ishikawa, H. Alternating Copolymerization of Methyl α -Phenylacrylate and Various Methacrylates. *Polym. J.* **1979**, *11* (4), 323-330.
- (85) Chikanishi, K.; Tsuruta, T. Reactivity of α -alkylacrylic esters. I. Homopolymerization behaviours of methyl α -alkylacrylates. *Makromol. Chem.* **1965**, *81* (1), 198-210.
- (86) Carneiro, O. S.; Covas, J. A.; Vergnes, B. Experimental and theoretical study of twin-screw extrusion of polypropylene. *J. Appl. Polym. Sci.* **2000**, *78* (7), 1419-1430.